
Chain-of-Thought Tokens Are Addressable Working Memory

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Abstract

Where do language models store the intermediate values used in multi-step reasoning? One hypothesis is that models reason in an internal workspace and externalize intermediate results as chain of thought only when that workspace becomes insufficient. We find no evidence for this capacity-triggered account. Across models from 1.5B to 671B parameters, accuracy on chained arithmetic collapses beyond one or two dependent steps when writing is forbidden, yet in free generation models write 93-100% of ground-truth intermediate values even at depth one, with no threshold at which externalization begins. None of four interventions induces the predicted shift between internal and written state: residual-stream lesions do not elicit more writing, token limits compress the trace but preserve intermediate values until truncation, filler tokens do not substitute for meaningful intermediates, and removing the selected active J-space concept subspace has no greater effect than removing size-matched random directions. We instead find that written token positions function as addressable working memory. Using causal interventions on six unmodified models from four families (4B to 14B), we replace the hidden state at a written number with the state produced by a counterfactual number while leaving the visible text unchanged. Among cases in which the model demonstrably relies on the written intermediate, this redirects the final answer to the counterfactual, never-written value in 70-100% of runs; size-matched random controls do so in at most 1% on the four instruction-tuned models. Blocking later tokens from attending to the written value removes the effect on all four models tested; in Qwen3-4B, retrieval is distributed across the middle third of the network and complete by layer 23 of 36. Two written positions store independently manipulable values that compose into a sum absent from the preceding context in 88-90% of runs. Together these results identify a write-store-retrieve-compose mechanism for serial reasoning: the written trace is part of the model's working-memory system rather than a record of it. They also expose a limit of text-only monitoring: under intervention, the same visible trace can carry different causally active states and produce different answers. Whether such text-state divergences arise naturally remains open.

1 Introduction

Where does a language model store an intermediate result after writing it? In multi-step reasoning the model computes a value, writes it into its chain of thought, and uses it several steps later. The value could persist inside the network, with the written token a transcript of computation that happened elsewhere, or the written position itself could hold it. For serial numeric reasoning we give a causal answer: written token positions function as addressable working memory. The hidden state at

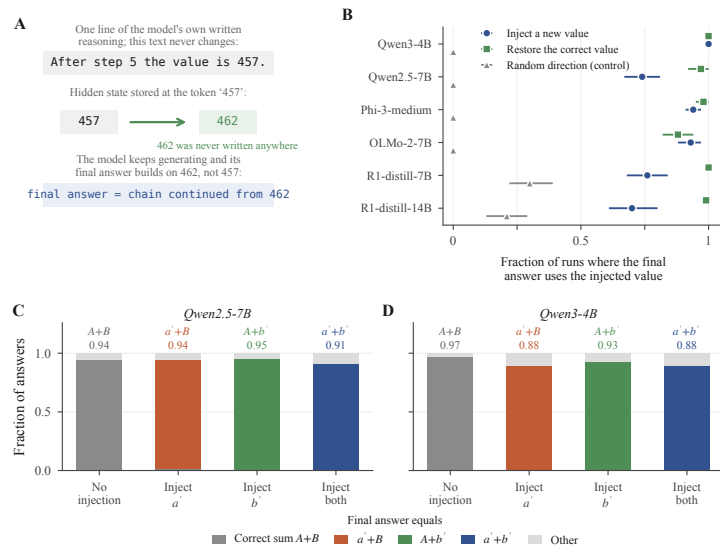


Figure 1: Same text, different stored value, different answer. (A) The model wrote “After step 5 the value is 457.” The text never changes; the hidden state at the 457 token is overwritten with the state a run that wrote 462 would have produced, and the final answer continues from 462, a value appearing nowhere in any text. (B) Fraction of runs whose final answer uses the injected value, per model: planting a counterfactual value (circles), restoring the correct value under corrupted text (squares), and a size-matched random control (triangles); bars are 95% bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, which raises their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

37 a written number stores causally active task state, later computation retrieves that state by attending
 38 to the number’s position, and multiple positions hold independently settable values that compose.

39 The central demonstration is concrete (Figure 1A). A model has written “After step 5 the value is
 40 457.” Every visible character stays in place; we overwrite the hidden state at the 457 token with
 41 the state from a run in which 462 was written there. The model continues its own reasoning and
 42 the final answer builds on 462, a number that appears nowhere in any text. Among items where the
 43 model demonstrably relies on the written intermediate, the planted value redirects the final answer
 44 in 0.70 to 1.00 of runs across six unmodified models from four families; size-matched random noise
 45 redirects at most 0.01 on the four instruction-tuned models (Section 4).

46 The experiments assemble into a four-part mechanism:

- 47 1. **Write** (behavioral, 1.5B to 671B). Models externalize intermediate values from the first
 48 step, at every scale, rather than only after an internal workspace fills; denied writing they
 49 fail beyond one to two dependent steps, and no intervention shifts state across the boundary
 50 (Section 3).
- 51 2. **Store** (causal, 4B to 14B). The hidden state at a written value contains causally active task
 52 state: restoring it corrects a corrupted trace, and planting a counterfactual value redirects
 53 the answer (Section 4).
- 54 3. **Retrieve** (causal, 4B to 14B). Later computation accesses the stored value by attending to
 55 its position; blocking that attention removes the dependence, and layer sweeps place the
 56 read in the middle third of the network (Section 5).
- 57 4. **Compose** (causal, 4B and 7B). Two positions behave as independently addressable slots
 58 whose planted values combine into a sum absent from all preceding text (Section 6).

59 J-space, a proposed global workspace identified by a Jacobian-based lens over residual activa-
 60 tions [Gurnee et al., 2026], is the contrast case. If serial reasoning state lived there, writing should

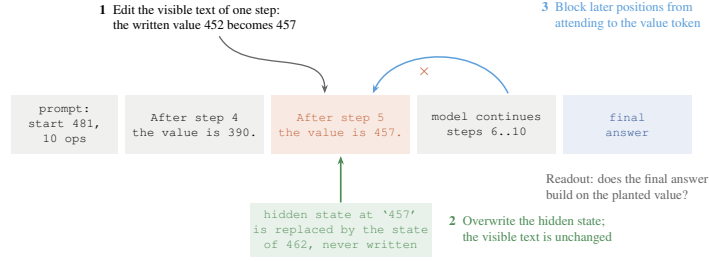


Figure 2: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

begin when the workspace fills and removing the workspace should impair reasoning. Neither happens: writing begins at step one, and removing the selected top-16 J-space concept subspace has no greater effect than removing matched random directions at any tested strength, while the planted value, which sits in the same residual states the lens reads, survives the removal. Causally active task state can be invisible to a fitted concept readout.

The mechanism has one consequence per audience. For reasoning, on these serial tasks writing is part of the computation rather than a report produced after it. For interpretability, a lens-negative result is not a state-negative result. For oversight, the visible trace does not uniquely determine the state retrieved from it; the divergences shown here are intervention-induced, and whether they arise naturally is untested. Section 7 bounds when models reuse rather than recompute written state. Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on program-variable tasks [Zhu et al., 2025]; the experiments here combine into a unified causal account of storage, retrieval, and composition in written serial reasoning on unmodified models.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events and surface forms. In both, the model writes one line per step giving the running value. A third task, box tracking, updates the contents of several containers in parallel; it appears only as the internally held comparison task for the lesion test in Section 3. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

2.2 Edit-and-continue protocol

The protocol takes a correct worked solution, changes the value at one step j , truncates immediately after the edited line, and lets the model continue (Figure 2). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95% bootstrap intervals over items. Appendix C summarizes generation and intervention configuration; prompts, model revisions, and per-run data will be released with the code.

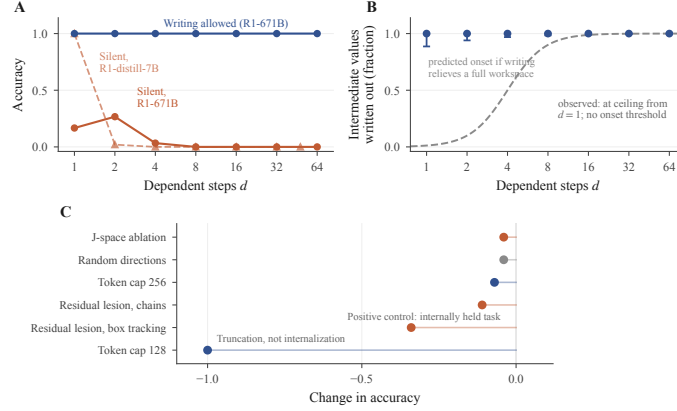


Figure 3: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); the dashed curve is the onset the capacity-relief account predicts, writing switching on past the internal failure depth, and it does not appear. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

3 Write: externalization is not overflow

3.1 Performance without written intermediates

Instructed to output only the final answer (Figure 3A), with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d=1$, 0.04 at $d=4$; DeepSeek-R1-671B: at most 0.27 at any depth, 0.03 at $d=4$; DeepSeek V3.2 on modular-arithmetic chains: 0.23 at $d=1$, at most 0.03 beyond). Prefilling the answer turn with inert dot tokens matched to the twelve-token-per-step budget of real traces [Pfau et al., 2024] does not rescue the collapse: at $d=4$, 0.13 with filler against 0.07 without on Qwen2.5-7B and 0.07 against 0.00 on Qwen3-4B, and 0.00 for both at $d=8$ ($n=30$ per cell). Qwen3-4B alone recovers at $d=2$ (0.40 to 0.97), the parallel-width benefit filler tokens are known to provide, which does not extend to the depths where writing is needed. The values are random, so nothing can be recalled; the failures bound performance under this instruction, with its prompting and training-distribution effects, rather than latent internal capacity.

3.2 When models write intermediate values

Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d=1$ through $d=64$, at every scale tested: no difficulty threshold at which writing begins is observed. This rules out an inference-time switch from internal holding to writing; an always-write policy learned in training could itself descend from capacity limits.

3.3 Tests for reallocation between internal and written state

A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held internally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step. Removing the selected top-16 J-space concept subspace at every position has no greater effect than removing matched random directions. At the calibrated coherence-safe dose (next-token agreement 0.87, perplexity ratio 1.12) it leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged on synthetic chains and on GSM8K. A dose sweep extends the null across the full tested range: from a quarter of the calibrated strength to complete removal of the selected subspace (α

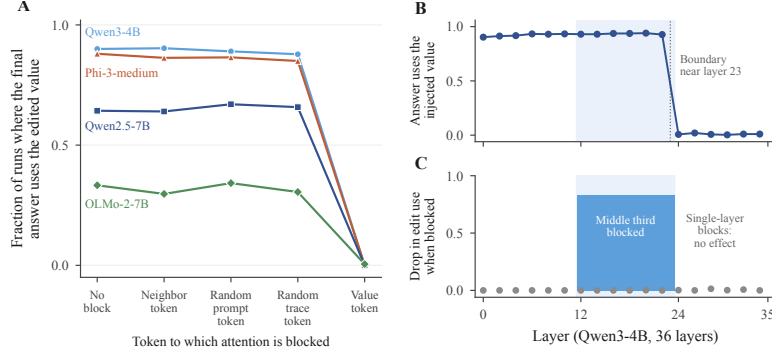


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing. (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking attention across the shaded middle band removes most edit use; no single layer’s block matters (gray dots).

from 0.125 to 1.0 at $k = 16$, layers 9 to 14 on Qwen2.5-7B), accuracy stays 0.99 to 1.00 in every condition on one- and two-step chains ($n = 80$ per cell). The chain-of-thought-versus-direct ablation asymmetry reported for J-space [Gurnee et al., 2026] therefore does not replicate on our tasks at any tested strength, including the one- to two-step depths where models do perform the arithmetic internally. This bounds the fitted readout and the selected active directions, on our tasks and doses, rather than all of J-space (Section 9); Section 4 gives the gap an existence proof: a value planted in the hidden state at a written number is causally decisive, sits in residual states the lens reads, and survives removal of the top 16 concepts.

4 Store: written positions hold causally active values

Figure 1 decomposes the effect of editing a written value. The rates are conditioned on the cases where the memory is demonstrably in use: items on which the model follows a visible text edit of the same intermediate (0.44 to 0.98 by model; Appendix A gives the per-model conditioning rates). Among these, restoring the correct hidden state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases, and planting a counterfactual value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. The patch replaces the position’s full contextual state rather than an isolated numeric feature, so the precise claim is one of sufficiency: the state at the written position is causally sufficient to make later computation behave as if a different value had been written. Because answers track whichever value’s state is planted, the value that state carries is the causal quantity, and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B). On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

5 Retrieve: attention to the written position

5.1 Attention-blocking experiments

Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s dependence on that value on every model tested (Figure 4A); matched blocks of the adjacent words, a random prompt position, or a random trace position leave it at the unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, a positive control showing the mask disrupts computation it targets. The collapse holds across three families and both task formats (natural-language: 0.730 to 0.007 and 0.720 to 0.003 on Qwen3-4B and Phi-3-medium). Under the block the

model produces a fluent, parseable number matching neither value (unparseable rate 0.000; OLMo-2-7B 0.040, its unblocked level): it computes forward from state it can no longer retrieve. Reading is the preferred path rather than the only one: with the value’s characters replaced by unreadable dots and attention untouched, the Qwen models reconstruct the value from the previous line in 0.60 to 0.73 of cases, Phi-3-medium never.

5.2 Where retrieval occurs across layers

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (Appendix B) show the same structure with model-specific depth.

6 Compose: independently addressable slots

In a task with two counters summed at the end (Figure 1C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.88 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.82). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The reported number must be assembled at answer time from two values set independently and never written anywhere.

7 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value whose derivation the prompt states is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly. Llama 3.3 70B follows edits at 0.97 either way, despite solving the same chains from scratch: the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. The conditions are separate experiments; we claim within-experiment contrasts only.

On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2: a benchmark intermediate is often one operation from numbers still in the prompt. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts offer a candidate explanation, that the direction depends on the state’s regenerability. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale. Trace reliance here is a separate quantity from verbalization faithfulness, whether the trace reports the computation that produced the answer; our claims concern reliance on already-written state.

8 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally. Wang [2026] corrupt surface traces only in aggregate and conclude reasoning is latent; our token-level edits show written state is causally decisive when the model cannot regenerate it, which bounds that conclusion. Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight channel; our results identify what such a monitor reads and what it cannot.

9 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and richer multi-entity state are untested. The J-space null bounds the fitted readout and the selected active directions, on our tasks and doses: the full-strength sweep covers six middle layers of one model on synthetic chains, and the lens’s domain excludes attention and MLP interiors, the KV cache, and residual directions outside the fitted dictionary. The fine layer sweeps cover one model, and the coarse sweeps do not resolve Qwen2.5-7B’s read boundary. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following makes cross-model levels not directly comparable (Appendix A). A structured patch control planting a valid activation for a different quantity would isolate value content beyond the size-matched noise control used here.

10 Conclusion

In serial numeric reasoning, the written trace functions as the model’s effective working memory. Models write intermediate values from the first step at every scale tested, the hidden state at each written value stores causally active task state, later computation retrieves it through middle-layer attention to the value’s position, and separate positions hold independently settable values that compose. Removing the fitted J-space readout of the same residual states leaves all of this intact, so causally active state can be invisible to a concept lens. For oversight, text is not a complete readout of the state attached to a reasoning token: under intervention, the same visible trace carries different causally active states and produces different answers [Korbak et al., 2025]. Our interventions establish this possibility; whether comparable text-state divergences arise naturally is unknown. Code, plans, and multi-entity state are the next test.

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A Per-model edit and patch rates

Table 1 gives the per-model rates behind Figure 1B, with the text-edit-following rate that conditions the patch columns. Ten-step chains; brackets are 95% bootstrap intervals; n counts items where the text edit was followed. Unconditioned per-run outcomes are in the released data. The distill models

re-solve the problem after their reasoning phase, which elevates their random-control reversion: in a text-logged sample, 24 of 24 reverting random-control continuations re-derive from the prompt’s starting value after the reasoning phase, and none correct the value in place. Re-solving can only produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is excluded (12 usable items; random control exceeds the planted-value rate).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

B Coarse cross-model layer sweeps

The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the fine sweep’s structure: the planted value takes effect at any band before the read and at none after. Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary is not resolved at this granularity.

C Generation and intervention configuration

Generation uses temperature 0.6, top- p 0.95, five samples per item. The text edit changes only the visible token. The state patch overwrites the residual stream at the value’s token position across a middle band of layers with activations captured from a run in which the different value was written there. The attention block masks all attention from post-edit positions to the value’s token at all layers. Sample sizes are 100 to 600 per condition with two to three seeds. The filler condition prefills the assistant turn with twelve inert dot tokens per dependent step, then decodes the answer at temperature 0. The J-space dose sweep runs the ablation of Section 3 at $k = 16$ and $\alpha \in \{0.125, 0.25, 0.5, 0.75, 1.0\}$ with identical items across arms. Prompts, model revisions, and per-run configuration files are released with the code and data.